Hemoglobin
Structure & Function

HMIM224

Objectives of the Lecture

1- Understanding the main ***structural & functional*** details of hemoglobin as one of the hemoproteins.

2- Identify **types** & relative concentrations of normal adult hemoglobin with reference to **HBA1c** with its clinical application.

3- Recognize some of the main genetic & biochemical aspects of ***methemoglobinopathies*** with some implications on clinical features (with focusing on ***thalassemias***).

Hemoglobin is a globular hemoprotein

- **Hemeproteins** are a group of specialized proteins that contain **heme** as a tightly bound prosthetic group.
- **Heme** is a complex of **protoporphyrin IX** and **ferrous iron (Fe2+) .**
- The **iron** is held in the center of the heme molecule by bonds to the four nitrogens of the porphyrin ring.
- The **heme Fe2+** can form two additional bonds, one on each side of the planar porphyrin ring.

  In myoglobin and hemoglobin, one of these positions is coordinated to the side chain of a **histidine residue** of the globin molecule, whereas the other position is available to bind **oxygen**

Globin of hemoglobin is a globular protein with a quaternary structure

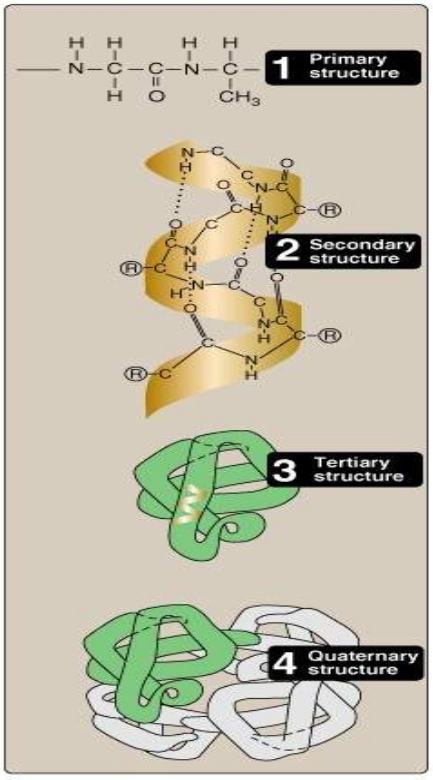


Figure 2.1

Four hierarchies of protein structure.

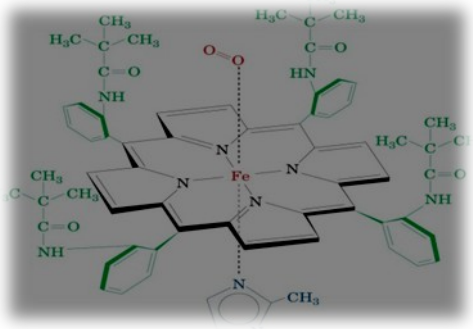
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Structure of heme

Heme is a complex of **protoporphyrin IX** and **ferrous iron (Fe2+)**.

The **iron** is held in the center of the heme molecule
by bonds of the four nitrogens of the protoporphrin ring.

Heme F2+ can form two additional bonds, one on each side of the porphyrin ring. One of these positions is coordinated to the Side chain of **histidine** residue of the globin molecule, whereas the other position is available to bind **oxygen**.



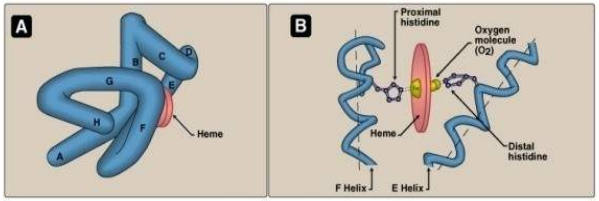


Figure 3.2

A. Model of myoglobin showing helices A to H. B. Schematic diagram of the oxygen-binding site of myoglobin.

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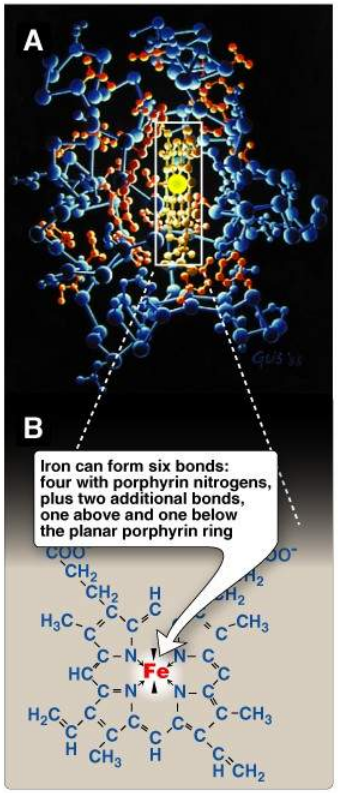


Figure 3.1

A. Hemeprotein (cytochrome c). B. Structure of heme.

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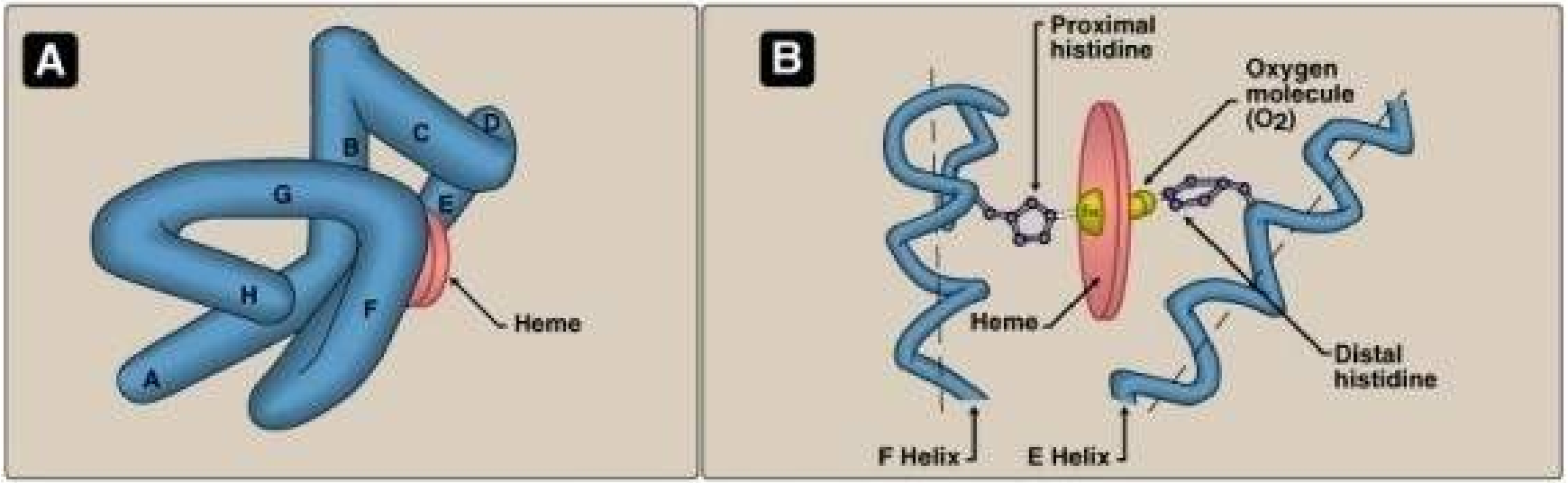


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A. Model of myoglobin showing helices A to H. B. Schematic diagram of the oxygen-binding site of myoglobin.

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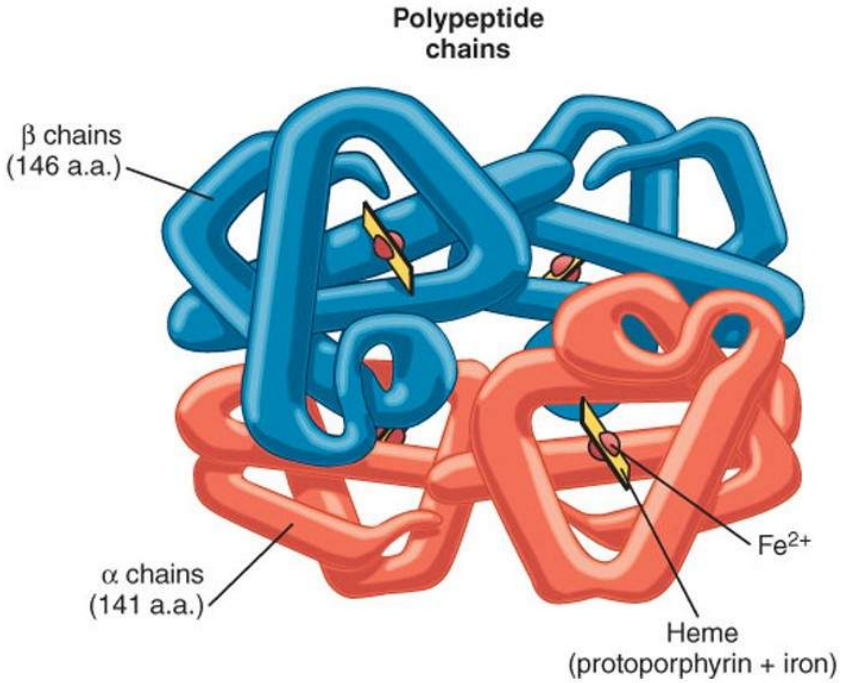
Structure & function of hemoglobin

- **Hemoglobin** is found exclusively in RBCs.
- Its main function is to transport oxygen from lungs to the tissues & carbon dioxide & hydrogen protons from tissues to lungs.

- **Hemoglobin A** is the major hemoglobin in adults, is composed of four polypeptide chains, 2 alpha (α) & 2 beta (β) chains, held together by noncovalent interactions

- **Each subunit** has stretches of α-helical structure & a heme binding pocket.

Structure & function of hemoglobin *(cont.)*

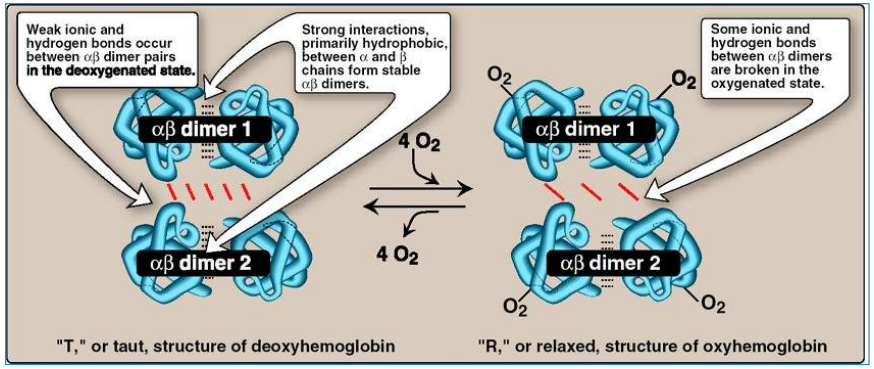


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Structure & function of hemoglobin *(cont.)*

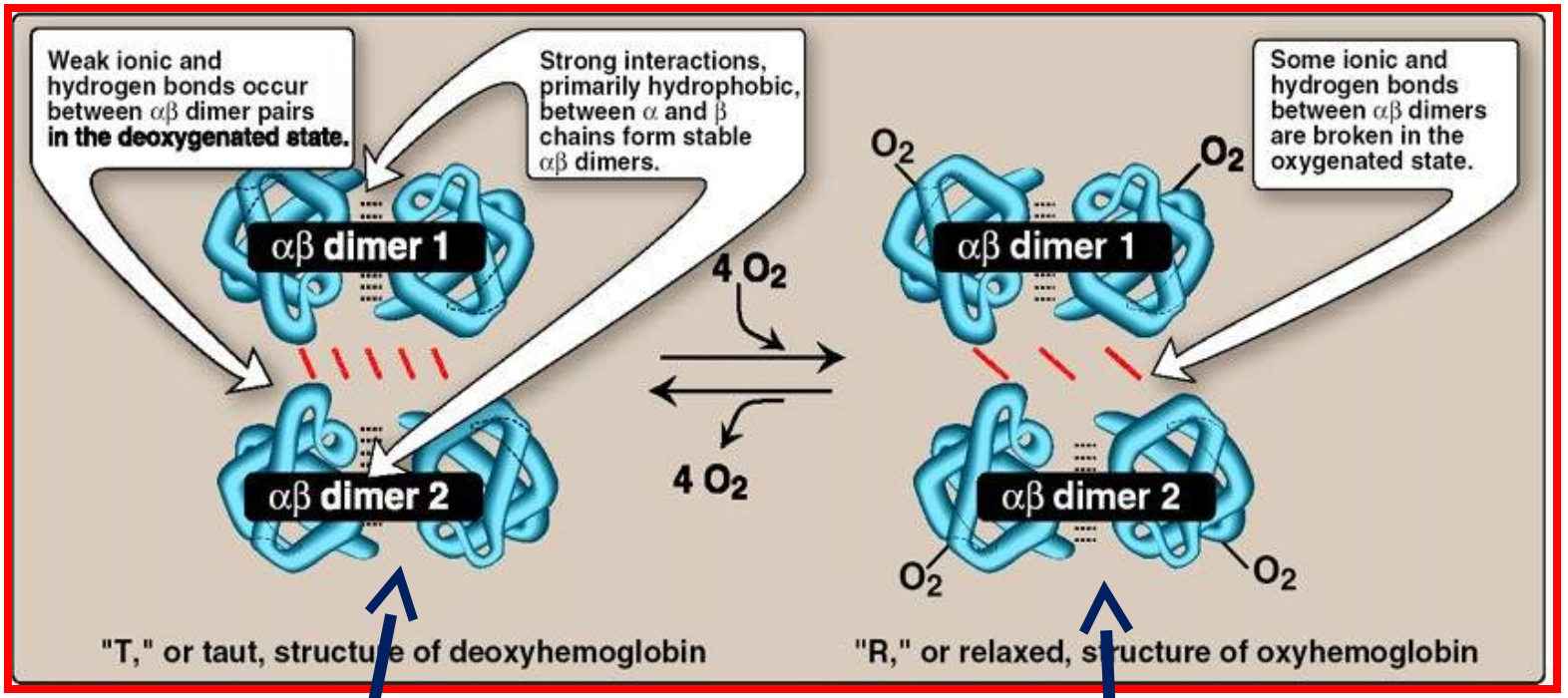
Quaternary structure of hemoglobin:

- The hemoglobin tetramer can be envisioned as being composed of two identical dimers, **(αβ)1** and **(αβ)2**, in which the numbers refer to dimers one and two.
- The two polypeptide chains within each dimer are tightly held together, primarily by ***hydrophobic interactions***
- In contrast, the two dimers are able to move with respect to each other, being held together primarily by ***polar bonds.***
- The weaker interactions between these mobile dimers result in the two dimers occupying different relative positions in **deoxyhemoglobin** as compared with **oxyhemoglobin**



oxygenation & deoxygenation of hemoglobin

(oxyhemoglobin & deoxyhemoglobin)



Deoxyhemoglobin
Taut structure

Oxyhemoglobin
Relaxed structure

Types of adult hemoglobin

Form	Chain composition	Fraction of total hemoglobin
HbA	$\alpha_2\beta_2$	90%
HbF	$\alpha_2\gamma_2$	<2%
HbA_2	$\alpha_2\delta_2$	2–5%
HbA_{1c}	$\alpha_2\beta_2$-glucose	% 6–3

HBA: the major hemoglobin in humans
HBA2: first appears 12 weeks after birth- a minor component of normal adult HB
HBF: normally synthesized only during fetal development
HBA1c : has glucose residues attached to β-globin chains – increased amounts in DM

Hemoglobin A1c (HBA1c)

Some of hemoglobin A is glycosylated
Extent of glycosylation depends on the plasma concentration of a particular .hexose (as glucose)

The most abundant form of glycosylated hemoglobin is **HBA1c** which has a glucose residues attached to β-globin chains in .hemoglobin RBCs

Increased amounts of **HBA1c** are found in RBCs of patients with diabetes mellitus .(DM)

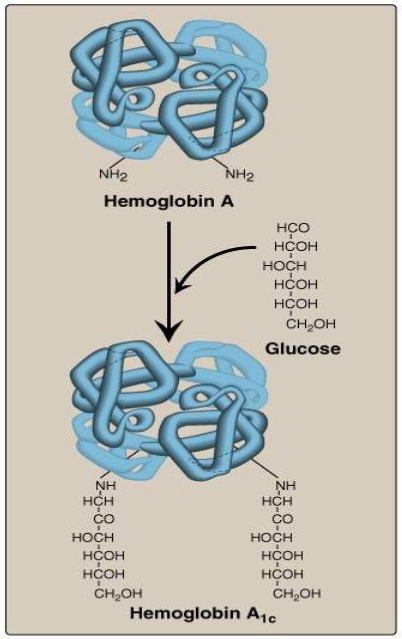


Figure 3.15
Nonenzymic addition of glucose to hemoglobin.

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HbA1c could be used as a monitor for the control of the blood glucose level during the last 2 months for diabetic patients

Hemoglobinopathies

Hemoglobinopathies are members of a family of genetic disorders caused by:

1- Production of a ***structurally abnormal*** hemoglobin molecule (Qualitative hemoglobinopathies)

Or: 2- Synthesis of ***insufficient quantities*** of normal hemoglobin (Quantitative hemoglobinopathies)

Or: 3- both (rare).

Thalassemias

- ***Thalassemias*** are hereditary hemolytic diseases in which an imbalance occurs in the synthesis of globin chains.
- They are most common ***single gene disorders*** in humans.
- ***Normally***, synthesis of α- and β- globin chains are coordinated, so that each α-globin chain has a β-globin chain partner.

  This leads to the formation of $\alpha_2\beta_2$ (HbA).
- ***In thalassemias***, the synthesis of either the α- or β-globin chain is defective.

Thalassemias *(cont.)*

Thalassemia can be caused by a variety of mutations, including:

1- ***Entire gene deletions (whole gene is absent)***

Or: 2- ***Substitutions or deletions of one or more nucleotides in the DNA.***

Each thalassemia can be classified as either:

1- A disorder in which **no globin** chains are produced
(α^{o}- or β^{o} -thalassemia)

Or: 2- Some β-chains are synthesized, but at **a reduced rate**.
(α^{+}- or β^{+}- thalassemia).

Thalassemias *(cont.)*

1- β-thalassemias:

- Synthesis of **β-globin** chains are decreased or absent, whereas α-globin chain synthesis is normal.
- α-globin chains cannot form stable tetramers, and therefore precipitate causing **premature death of RBCs**.
- Accumulation of $\alpha_2\gamma_2$ (**HbF**), γ_4 (**Hb Bart's**) & **α-chain precipitate** occurs.
- These factors end result in development of **chronic anemia** (hemolytic).

Thalassemias *(cont.)*

Some genetic aspects of thalassemia:

There are only two copies of the β -globin gene in each cell (one on each chromosome 11).

So, individuals with β -globin gene defects have ***either***:

1- ***β-thalassemia minor (β-thalassemia trait):***
if they have only **one** defective β-globin gene.

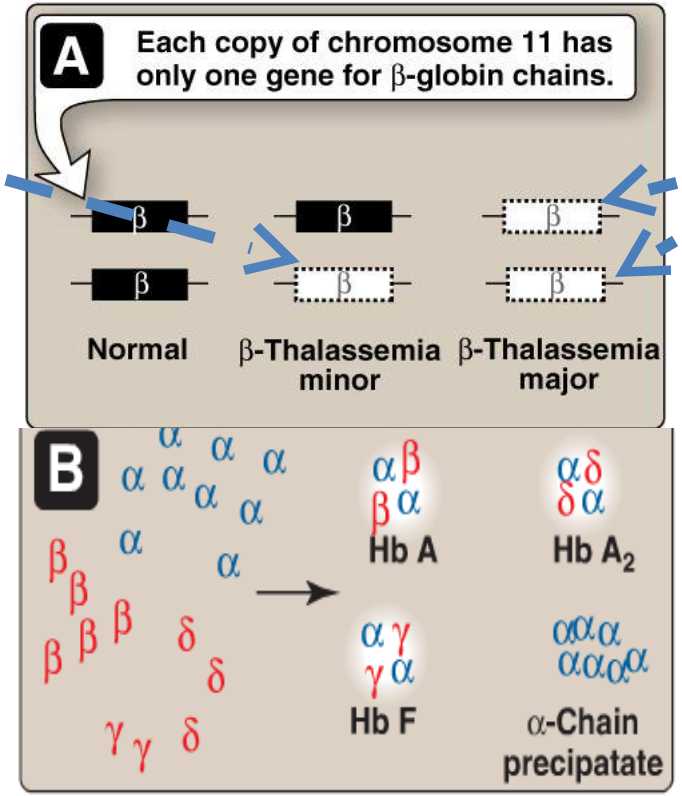
2- ***β- thalassemia major (Cooley anemia):***
if **both** genes are defective.

Thalassemias *(cont.)*

Mutation in one of β-globin genes

↓

β-thalassemia minor



Mutation in both β-globin genes

↓

β-thalassemia major

Figure 3.22

A. β-Globin gene mutations in the β-thalassemias. B. Hemoglobin tetramers formed in β-thalassemias.

Thalassemias *(cont.)*

Some clinical aspects of thalassemias:

1- As β-globin gene is not expressed until late fetal gestation, the physical manifestations of ***β- thalassemias*** appear only after birth.

2- Individuals with ***β- thalassemias minor***, make some β-chains, and usually require no specific treatment.

3- Infants born with ***β- thalassemias major*** seem healthy at birth, but become severely anemic during the first or second years of life.
They require regular transfusions of blood.

In these cases, bone marrow replacement therapy is recommended.

Thalassemias *(cont.)*

2- α-thalassemia:

Synthesis of α-*globin chains is decreased or absent.*

Each individual's genome contains four copies of the α-globin (two on each chromosome 16), there are several levels of α-globin chain deficiencies

Thalassemias *(cont.)*

Types:

If one of the four genes is defective
the individual is termed a **silent carrier of α- thalassemia** as no physical manifestations of the disease occur.

If two α-globin genes are defective,
the individual is designated as having **α-thalassemia trait**.

If three α-globin genes are defective;
the individual has **hemoglobin H** (**HbH**) disease which is a mildly to moderately hemolytic anemia.
Synthesis of unaffected γ- and then β- globin chains continues, resulting in the accumulation of γ tetramer in the newborn (γ_4, **Hb Bart's**) or β-tetramers (β_4, **HbH**).
The subunits do not show heme-heme interactions. So, they have very high oxygen affinities. Thus, they are essentially useless as oxygen carriers to tissues

If four α-globin genes are defective,
hydrops fetalis & fetal death (death at birth), occurs as α-globin chains are required for the synthesis of **HbF**

Thalassemias *(cont.)*

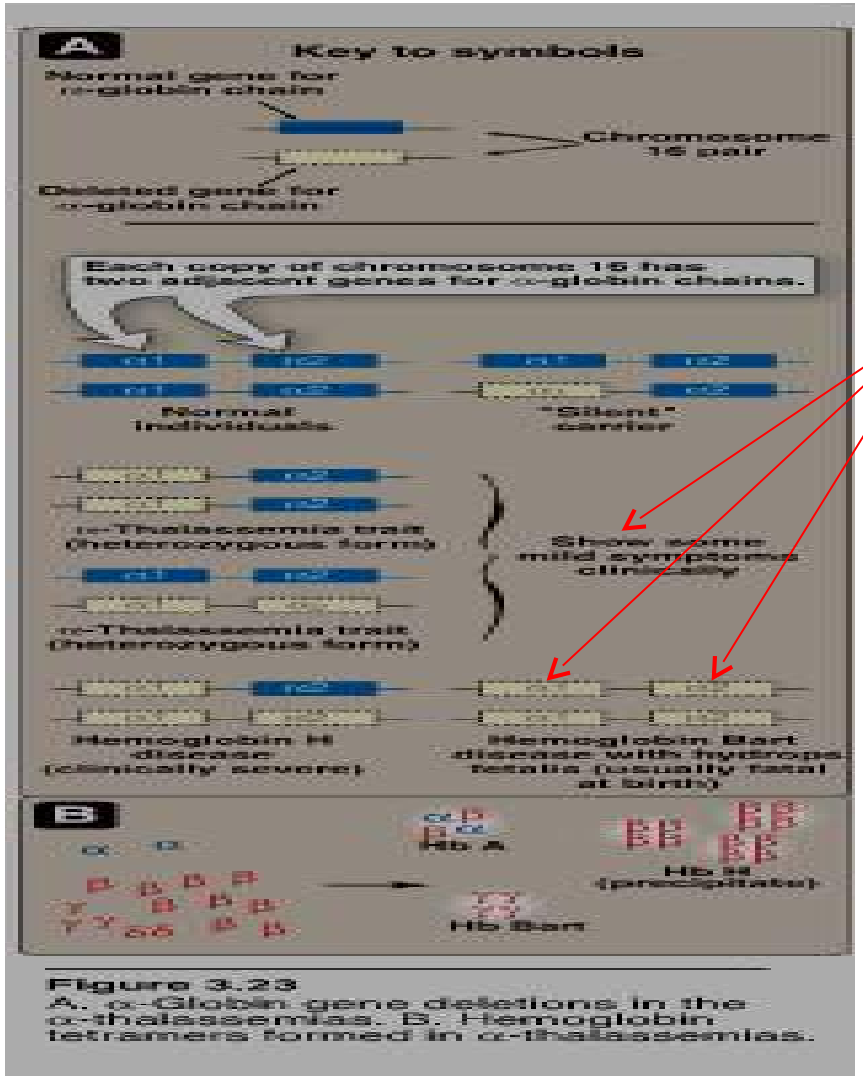


Figure 3.23
A, α-Globin gene deletions in the α-thalassemias. B, Hemoglobin tetramers formed in α-thalassemias.

Types of α-thalassemias

Assignments

- **Methemoglobin**
- **Sickle cell anemia (Genetic, Biochemical & Clinical Aspects)**